

Client

Browser client

Public landing page + authenticated app views

Landing page
Public, no login
Login / Register CTA**Upload + convert**
Drag & drop
Preview, reorder**My PDFs**
History list
Search by title
Rename / delete**Auth views**
Login
Register**Download**
PDF stream
to browser

Controllers

ASP.NET Core 8 — controllers

Thin HTTP handlers; delegate all logic to services; [Authorize] on protected routes

HomeController
Landing page
Public route**UploadController**
[Authorize]
Upload, validate
IFormFile**PdfController**
[Authorize]
CRUD + search
Convert, download**AccountController**
.NET Identity
Register, Login, Logout

Services

Service layer

All business logic; injected via DI; controllers stay thin

ImageValidator
Type, size, count
Error messages**PdfGenerator**
PdfLibrary wrapper
Orientation, embed**TempFileService**
Save, retrieve
Clean up on expiry**PdfManagement**
CRUD on SavedPdf
Search by title

Storage

Storage

SQL Server / SQLite via EF Core + server file system

Identity DB (users)

App DB (SavedPdf, Session)

File system (PDFs, temps)